

 <https://github.com/ColitoBandito/cs4412-project-Collin>

Mapping Mood Profiles and Feature Associations of Spotify Tracks

April 3, 2026

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Data Mining W01

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KEY EDA INSIGHTS

This initial phase of exploratory analysis has shown a number of patterns that give an idea how to approach the mining phase of this project.

- **Strong Feature Correlations:** A high positive correlation (0.77) was found between Loudness and Energy, while there was a significant negative correlation (-0.71) between acousticness and energy.
- **Data Density:** Early scatter plots showed that Valence and energy had a dense, overlapping distribution across all features. This confirmed that algorithmic clustering is helpful in distinguishing "mood profiles", beyond just regular listener intuition.
- **Feature Skew:** Attributes such as Instrumentalness skewed strongly towards zero, which indicates that, at least within this dataset, the majority of tracks were vocal-centric tracks.

Major Preprocessing Decisions

Data integrity was paramount in ensuring proper behavior of distance-based mining algorithms. Two primary transformations were applied.

- **Standardization:** Some attributes such as Tempo (BPM) and Loudness (db) existed on much larger scales than others, with Valence being only on a scale of (0.0-1.0). As such all numerical attributes were standardized using StandardScaler to eliminate any potential magnitude bias.
- **Dimensionality Reduction:** Principal Component Analysis was utilized to reduce the amount of high dimensional space into two primary components. This was to enable visualization of clusters on a 2D plane without sacrificing maximum variance.

MINING RESULTS AND INTERPRETATION

The Elbow Method was used to determine $k = 4$ as the optimal number of clusters. The rate of decrease in inertia leveled off substantially at this point. Four "Mood Profiles" were discovered. Each Profile is identified by its mathematical center:

Table 1: Discovered Spotify Mood Profiles ($k = 4$)

Cluster	Key Characteristics (Centroids)	Proposed Mood Label
0	High Valence (0.68), High Danceability (0.70)	Party
1	High Acousticness (0.84), Low Energy (0.20)	Acoustic
2	High Energy (0.67), High Loudness (-7.51 dB)	Dark
3	Highest Speechiness (0.36), High Danceability (0.68)	Lyrical

- Cluster 0 (Party): Representative of positive sounding music. High valence suggests the music here contains an optimistic tone, that it's easy dance to and reminiscent of modern Pop.
- Cluster 1 (Acoustic): The most distinct of all profiles generated, it shows a strong inverse relationship in acousticness and energy. Representative of tracks that are "unplugged" and lower tempo.
- Cluster 2 (Dark): Energetic but not happy music sonically, this grouping contained music with more serious tones. It includes most tracks where high intensity isn't associated with a positive mood.
- Cluster 3 (Lyrical): Contains tracks that are on average 5x more wordy than the other groups. Groups music that is high in wordiness, which encompasses rap/hip hop that is performed on top of instrumentation.

VALIDATION/SILHOUETTE ANALYSIS

While the Elbow Method suggested an inflection point at $k = 4$, a Silhouette Analysis was performed to mathematically validate this choice.

- Validation: The Silhouette Score was highest at $k = 2$ (0.336), indicating a broad split between acoustic and electronic tracks. However, $k = 4$ (0.181) was selected as the "Sweet Spot," as it provided significantly more analytical depth than $k = 2$ while maintaining better separation than $k = 5$ (0.176).

M3 EXPANSION: DECISION TREE DISCOVERY

To move beyond the visual limitations of PCA (which captured 55.1% of variance), a Decision Tree Classifier was implemented to audit the K-Means clusters in their original 11-dimensional space. This served as the primary analytical expansion for this milestone.

- **Rule Discovery:** The tree identified Valence (≤ 0.47) as the root decision node, followed by Energy and Speechiness.
- **Cluster Validation:** The tree successfully "mapped" the logic of our mood profiles, proving that high Energy distinguishes "Dark" tracks from "Acoustic" ones, while high Speechiness (> 0.26) is the definitive marker for the "Lyrical/Hip-Hop" segment.
- **Synthesis:** This expansion confirms that our unsupervised clusters are grounded in clear, hierarchical audio rules that were previously obscured in the 2D PCA plot.

MILESTONE 4 PLAN

The final phase of this project will focus on technical refinement and portfolio-ready presentation.

- **Comparative Mining:** I will implement DBSCAN to identify niche, density-based sub-genres and compare them against the current K-Means profiles.
- **Validation:** A final assessment of model limitations and a deeper investigation into "Anomalous Track Profiles" identified by the clustering logic.
- **Deliverables:** Finalizing the GitHub repository with a professional-grade README and cleaning all notebook documentation for public viewing.